that ecosystem will change dramatically. The importance of keystone species was brought to light by the American biologist Robert Paine—who derived the term from the central “keystone” at the top of an arch that stops it from collapsing—in his 1969 article “A Note on Trophic Complexity and Community Stability.”

“Do you want an auto mechanic who…can name, list, and count all of the parts of your engine, or one who really understands how each part interacts with the others to make a working engine?”

Robert Paine

The keystone concept

In the 1960s, Paine spent several years studying the animals of the intertidal zone of Tatoosh Island on the Pacific coast of Washington State. He removed the ocher starfish and watched its key prey, a mussel whose numbers had been kept in check by the starfish, dominate the zone, replacing other subordinate species. The removal of a single, keystone species had a clear impact on many others. Paine developed the idea to include the concept of “trophic cascades”—the strong, top-down effects that ripple through an ecosystem and its organisms. Since Paine’s work with starfish, several studies have demonstrated that there are many other keystone organisms, and they each fulfill their role in different ways.